GEOMETRIC CONNECTEDNESS IS EQUIVALENT TO CONNECTEDNESS OF THE BASE CHANGE TO A SEPARABLE ALGEBRAIC CLOSURE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-varieties/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup, of the native theorem declarations with mathematical macros, and of the project's own no-op definitions of the reference display environments; no macro was invented or redefined.

Editorial note. The same-chapter referenced result is located at its \label line inside the pinned chapter file.

Allowed standard prerequisites (editorial).

- Geometrically connected scheme over a field. Same chapter, the definition introducing geometrically connected schemes by connectedness of the base change to an algebraic closure. Supplies the notion being characterised; the reduction from a separable algebraic closure to an arbitrary extension is the counted content.
- Connectedness over a separably closed field may be tested after an arbitrary extension of that field. Same chapter, Lemma labelled lemma-separably-closed-field-connected-components (source line 771, Stacks tag 0363); invoked at proof lines 805-806. Supplies that X over the separably closed field kbar stays connected after extending scalars to kbar'; the surjectivity argument that returns to k' is retained in the counted span.

Source reference. Original lemma 0363: sources/stacks-varieties/pinned-original-source.tex, source lines 771–771. Same-chapter pointer invoked at proof lines 805-806 for the case of a separably closed field; the choice of the comparison field kbar' and the surjectivity descent to X_{k'} are retained verbatim in the counted span.

Lemma 0.1. *Let k be a field. Let X be a scheme over k. Let $\overline{k}$ be a separable algebraic closure of k. Then X is geometrically connected if and only if the base change $X_{\overline{k}}$ is connected.*

Proof. Assume $X_{\overline{k}}$ is connected. Let k'/k be a field extension. There exists a field extension $\overline{k}'/\overline{k}$ such that k' embeds into $\overline{k}'$ as an extension of k. By Lemma 0363 we see that $X_{\overline{k}'}$ is connected. Since $X_{\overline{k}'} \to X_{k'}$ is surjective we conclude that $X_{k'}$ is connected as desired. □

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.